Roadmap: from chip to an embodied, closed-loop fly

What NeuroMPS builds next, after the final round. The plan below is a commitment, not a result: nothing in it has been run yet except where a number cites `outputs/metrics.json`. Everything it will produce is simulated, for research decision support; none of it is a claim about any compound's effect in any animal or person.

Slide message: *Today we read a simulated fly brain as a chip and as an animal, through the neurons that drive behaviour. Next we give it a body and close the loop, so that the question "does the chip flag a drug before behaviour fails?" can be asked of movement itself, with sensory feedback in place. Each stage has a registered go/no-go test.*

1. Why close the loop

The chip-to-animal study (report Act VII) reads behaviour as the firing of named neurons, down to the motor neurons of the male nerve cord. Three things it cannot see:

- **Movement.** A motor neuron firing faster is not a leg moving differently. Muscles, joints and contact with the ground turn firing into behaviour, and they can mask or amplify a change.
- **Feedback.** In a real fly, movement is sensed and fed back: leg proprioceptors report joint angles and loads to the nerve cord and brain. A drug that disturbs a circuit inside a loop can have effects an open-loop model misses, in either direction.
- **The environment.** Feeding is a sequence (find, taste, extend the proboscis), not a firing rate; grooming is a sequence of leg sweeps.

Our current results are the baseline the embodied model has to be compared against. On the simulated fly brain, the chip's firing rate ranks drugs by how much they disrupt behaviour (Spearman 0.863), but it raises the alarm late. At the motor neurons of the male nerve cord, the chip is later than motor output in 4 drug families, level in 1, and never earlier. And electrodes on the brain's output tract caught GABA block in time where a planar array did not (0.361 against 0.044 log2). The embodied model asks whether these conclusions survive movement and feedback.

2. What already exists

Layer	Status	Where
Whole-brain model (FlyWire), drug knobs by transmitter class	built, validated against the published sugar→MN9 result	`src/neuromps/fly/model.py`; report §23
Brain and nerve cord (MaleCNS, 165,122 neurons)	built; replicates the behaviour results	`fly/malecns.py`; §25
Virtual MEA scored by the chip's own extractor	built	`fly/electrodes.py`, `features/mea.py`
Motor-neuron readouts (proboscis, front legs, all motor neurons)	built, registered study done	`fly/motor.py`; §25
Neuron-to-muscle map	in the data: 260 of 381 leg motor neurons are typed by the muscle they drive	`fly/roadmap_inventory.py`
Feedback neurons	in the data: 1,030 proprioceptive sensory neurons in the nerve cord, among them 403 chordotonal-organ and 175 campaniform-sensilla neurons	`fly/roadmap_inventory.py`
Body and physics	available, not yet used: NeuroMechFly v2 (FlyGym) and flybody, both on MuJoCo, all Apache-2.0	external

3. The loop we will close

1. **Brain and nerve cord** (MaleCNS, leaky integrate-and-fire, the drug knobs as now).
2. **Motor neurons to muscles.** Each leg motor neuron typed by its muscle drives that muscle's activation; the untyped ones are mapped by their connectivity to typed partners, or left out and reported.
3. **Muscles to joints.** Muscle activations become joint torques in the body model (NeuroMechFly v2's leg joints).
4. **Body and world.** MuJoCo integrates the body against the ground, with leg adhesion, and against a sugar spot for feeding.
5. **Joints back to neurons.** Joint angles and contact forces drive the matching proprioceptive sensory neurons (chordotonal organs for angles, campaniform sensilla for load), which re-enter the network.

The chip side does not change: the same virtual array, scored by the same extractor, read while the fly behaves.

4. Stages, each with a registered go/no-go test

Stage	When	Deliverable	Go/no-go test (registered before it runs)
1. Open-loop legs	weeks 1–3	front-leg motor neurons drive NeuroMechFly's front legs	antennal drive produces front-leg sweeps toward the head that untreated, undriven runs do not
2. Feeding and walking	weeks 3–6	proboscis extension from the proboscis pool; walking from descending commands	sugar drives proboscis extension and bitter suppresses it; a walking command produces an alternating gait
3. Close the loop	weeks 6–10	proprioceptive feedback into the nerve cord	feedback measurably changes the movement (it stabilises or reshapes the rhythm) against the open-loop run
4. Drugs in the loop	weeks 10–14	the registered chip-to-animal design rerun with movement as the readout	registered predictions, among them: does the chip still flag drugs after movement fails; does the output-tract electrode still lead
5. Check against real flies	weeks 14–18	comparison with published fly pharmacology (feeding and grooming under GABA-A and GluCl drugs)	the direction of each published behavioural change is reproduced, or the mismatch is reported
6. Release	weeks 18–20	open code, models and a paper	one command reproduces every number, as now

Each stage stops if its test fails, and the failure is reported, as every failed prediction in this project has been.

5. What it would change for organ-on-a-chip

The embodied model lets three chip-design questions be asked of behaviour itself rather than of neurons:

- **Sensitivity.** How small a firing-rate change must a chip resolve to flag each mechanism before movement fails? Today's answer comes from interneurons and motor neurons (report §24).
- **Electrode placement.** Does recording the connection between compartments (the output tract, a chip's microchannels) still lead a planar array when behaviour is movement under feedback?
- **Screen thresholds.** How does a regulatory screen's hit cutoff compare with the change at which movement, not firing, fails?

6. Risks, and how each is handled

Risk	Handling
Part of the neuron-to-muscle map is missing (motor neurons not typed by muscle)	map the rest by connectivity to typed partners; report coverage; test robustness by leaving the uncertain ones out
A leaky integrate-and-fire nerve cord may not generate rhythms (walking needs them)	stage 2's go/no-go test catches it; then add the intrinsic currents rhythm generation needs, as a stated change
The male nerve cord is over-excitable under the published weight (any drive recruits many motor neurons)	calibrate one weight against the untreated motor recruitment before stage 4, registered
Real fly pharmacology is sparse and in different units	stage 5 compares directions, not magnitudes, and says which comparisons were impossible
Compute	FlyGym 2 simulates on GPU; the GB10 runs the network and the body in parallel

7. Status, and what stage 1 changed

Stage 1 ran and failed its go/no-go test, as did its registered follow-up. The male-CNS network was wired to the FlyMimic leg (MuJoCo 3.16, the leg's 15 Hill-type muscles driven by 49 of the 68 left front-leg motor neurons, by name). Antennal drive moved the claw toward the head in every trial, but sugar drive moved it closer still (median closest approach 0.9386 against 1.1067 mm): the movement was not specific. Stage 1b rescaled the male-CNS weight by the ratio of synaptic input per neuron between the connectomes; at that weight sugar no longer drove the feeding motor neuron at all (0.0 Hz), so its leg results are uninterpretable, as registered.

What went wrong. The same sugar drive puts 251 neurons above 10 Hz in the published FlyWire model and 14,263 in the male CNS at the same weight, most of them central-brain neurons (11,082 of 32,160); at the rescaled weight still 7,586. The published model's parameters do not carry over to the second connectome's brain: there, activity avalanches, and lowering one global weight silences weak pathways before it stops the avalanche. This is itself a result for the paper, and it changes the plan.

The revised plan. Use each connectome where its model is validated:

- **Stage 1c (next):** drive the male nerve cord from its descending neurons alone, without the male brain. Activating the descending neurons the literature ties to antennal grooming should move the front leg toward the head; descending neurons for other behaviours should not. This tests the nerve cord, the muscle map and the leg without the brain's avalanche.
- **Then:** join the FlyWire brain, where the published model is validated, to the male nerve cord through descending neurons matched by cell type across the two connectomes.

Both steps are registered before they run, like every stage.

Stage 1c ran: three of five registered predictions pass. With the male brain removed and named descending neurons driven, the nerve cord is silent without drive, and the descending neurons the literature ties to antennal grooming (aDN1, aDN2) drive the left front-leg motor pool at 15.846 Hz, against 2.72 for the middle and 1.697 for the hind leg; no other descending neuron recruits the front leg as strongly (forward walking 3.601 Hz). The kinematic prediction failed: grooming drive brings the claw to 1.1345 mm of the antenna, but forward-walking drive reaches 1.154 mm at the low points of a rhythmic swing, so the closest approach does not separate the two. The figure shows the difference the readout missed: under grooming drive the claw goes to the head and stays there; under forward-walking drive it oscillates. That is the readout to register next (stage 1d, fresh seeds), not a reason to re-score this one.

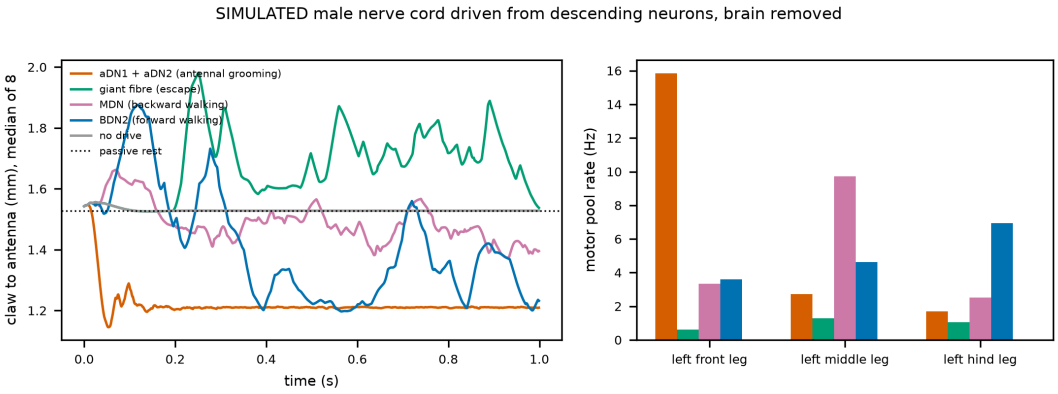


Figure 1. *The SIMULATED male nerve cord, brain removed, driven from descending neurons: claw-to-antenna distance of the FlyMimic leg, and the motor pools of the three left legs, per drive.*

Stage 1d: all four registered predictions pass, on fresh seeds. Read as the time the claw spends near the head and as the rhythmicity of its movement, the two drives separate cleanly: under grooming drive the claw is near the head for 0.9665 of the second, under forward-walking drive for 0.368, and never under the giant fibre or without drive; forward walking moves the leg more rhythmically (0.4599 against 0.2374); the result holds at every activation scale, and stage 1c's front-leg specificity replicates (front-leg pool 16.066 Hz under grooming drive). Rhythmicity alone does not mark walking: the giant fibre scores as high (0.4561), while keeping the claw away from the head. (Caveat found in stage 2e, below: this rhythmicity measure does not require an autocorrelation peak, so smooth movements without a rhythm also score high. The registered audit, below, re-scored D2 with a measure that requires a peak: forward walking drops to 0.0956 and the giant fibre to 0.1896, while grooming keeps 0.2374, so D2 fails under the corrected measure. The walking "swing" was an artefact; D1, the hold at the head, does not depend on the measure and stands.) **Stage 1's question is answered for the nerve cord:** the descending neurons that start antennal grooming in the fly move this leg to the head and hold it there, specifically. Next is the brain: the FlyWire brain joined to this nerve cord through type-matched descending neurons, so that sensory drive, not a stimulated command neuron, starts the movement.

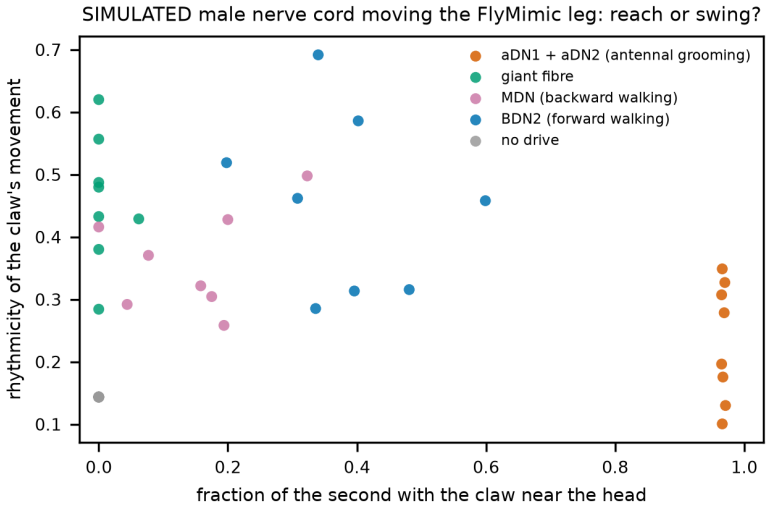


Figure 2. *Time near the head against rhythmicity, one point per SIMULATED trial: the antennal-grooming drive holds the leg at the head; walking and escape drives move it away (the vertical axis is the registered rhythmicity, found by the audit to score smooth movements high).*

Stages 1e and 1f: the FlyWire brain drives the male nerve cord, end to end. Each FlyWire descending neuron was paired with the male-CNS descending neuron of the same cell type and side (1,220 of 1,303 matched), the published brain model was run under antennal, sugar or no input, and its descending spikes were replayed into the male nerve cord, which moved the leg. Stage 1e's registered precondition, an absolute 10 Hz line for the grooming descending neurons, failed at 8.531 Hz, so its leg results are uninterpretable; it also showed that its time-near-head readout counted a mild flexion under sugar input as a reach. Stage 1f, registered with a relative precondition and the depth of reach as readout and run on fresh seeds, passes all five predictions. Antennal input brings the claw to 0.5369 mm of the antenna, sugar input to 0.9377 mm, no input leaves it at rest; the front-leg pool carries the movement; and removing the spikes of the two grooming descending neuron types alone (aDN1, aDN2; four neurons) returns the claw to 1.4793 mm. **Stage 1 is complete:** sensory input to a validated whole-brain model, carried by the descending neurons the fly uses for antennal grooming, moves a muscle-driven leg to the head; sugar input does not, and without those four neurons the movement is gone. Next is stage 2 (feeding and walking) and stage 3 (closing the loop).

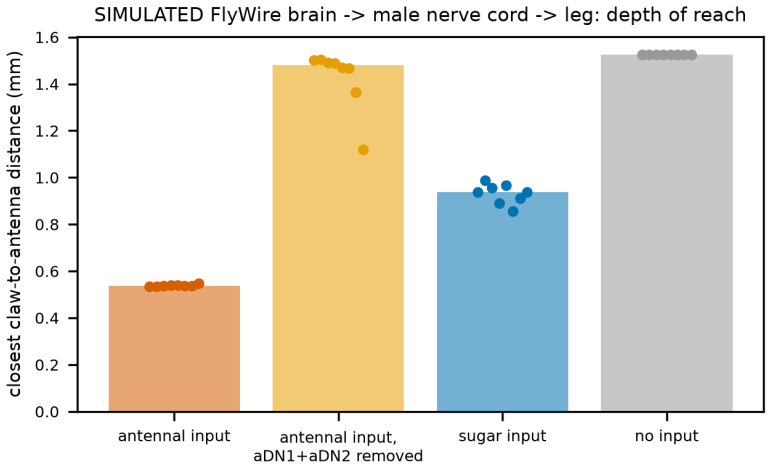


Figure 3. *Depth of reach per condition in the SIMULATED relay (FlyWire brain -> matched descending neurons -> male nerve cord -> FlyMimic leg), one point per trial: antennal input reaches deepest; removing aDN1 and aDN2 abolishes the reach; sugar input flexes the leg only partway.*

Stage 2a: no walking rhythm from the nerve cord alone (failed as registered). Before any body, stage 2a asked whether the male nerve cord turns its forward-walking command (BDN2, DNg100) into a motor rhythm: tibia flexors and extensors alternating within a leg, the two front legs alternating, and a tripod across legs, read from the twelve tibia motor pools of the six legs. Without drive the cord is silent (W0 passes). Under the walking command it is active but tonic: the left middle-leg tibia extensors fire at 54.12 Hz and the right hind-leg extensors at 34.12 Hz, while the tibia flexor pools of five of the six legs stay silent (left front flexors 0.0 Hz). The left front flexor pool has no rhythm (median rhythmicity 0.0), so no phase can be measured and W1 to W4 fail. Backward-walking drive (MDN) gives the same picture with different legs. This is the risk §6 names: a leaky integrate-and-fire nerve cord, with connectome weights and no intrinsic currents, has no half-centre oscillator. The next step is the one §6 commits to, a stated model change registered before it runs: add the intrinsic property rhythm generation needs (spike-frequency adaptation) and test again on fresh seeds, with the connectome-only result kept as the reference.

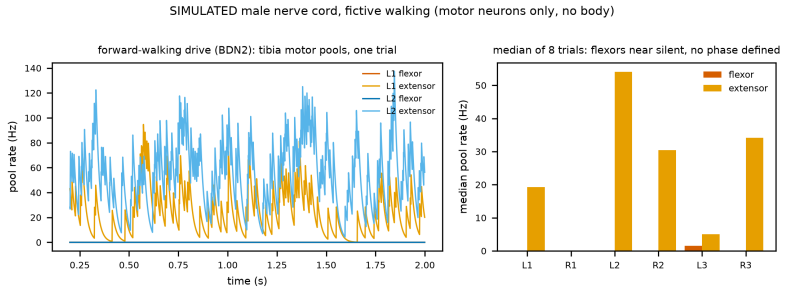


Figure 4. *Tibia flexor and extensor pool rates on the six legs under the walking command, SIMULATED fictive walking in the male nerve cord: extensors fire tonically, flexors stay silent, and there is no alternation.*

Stage 2b: sugar input to the FlyWire brain extends a simulated proboscis; bitter input gates it (six of six registered predictions pass). Every proboscis motor neuron that McKellar et al. (2020) tie to a muscle has a FlyWire cell type (through the male-CNS cross-reference). The rostrum protractor (MN9) and its retractors (MN1, MN2) set the rostrum's command; the haustellum extensors (MN4) and flexors (MN3) set the haustellum's. Both commands drive the proboscis of the flybody whole-fly model. Under sugar input the extending pools reach activations of 0.9644 (rostrum) and 0.8858 (haustellum), their antagonists 0.0561 and 0.0, and the proboscis extends fully (0.3431 mm peak; full extension is 0.3428 mm). Bitter input alone, antennal input and no input leave it at rest. Adding bitter input to sugar input halves the peak (0.1618 mm) and cuts the time-averaged extension from 0.3054 to 0.0488 mm: the proboscis starts to extend, then is held back to brief flicks. The result holds at the other activation scales. Only the MN9 part of this was seen before registration (it is the published model's result); the haustellum pools and their antagonists were not. Exploratory, not scored: sugar input recruits only feeding motor neurons (neck, eye, antennal and crop motor neurons stay silent), and the bitter gate acts on the proboscis pools (other proboscis motor neurons 52.25 Hz under sugar, 2.75 with bitter added) far more than on the ingestion pool (24.58 and 19.98 Hz). **Stage 2's feeding test passes; its walking test (2a) does not.** The body is held fixed and its joints are position servos, so this is the motor command made visible, not a muscle-level proboscis.

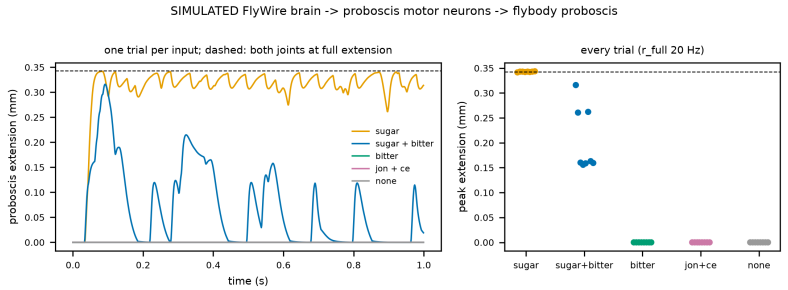


Figure 5. *Proboscis extension in the SIMULATED FlyWire brain -> proboscis motor neurons -> flybody chain: sugar extends the proboscis fully and holds it; adding bitter input lets it start and then holds it back; bitter, antennal or no input leave it at rest.*

Stage 2c: adding spike-frequency adaptation does not make the nerve cord walk (stopped as registered). §6 committed to adding an intrinsic property if stage 2a failed. Adaptation's two parameters have no published fly value, so they were swept over a registered grid of 36 cells. The registration fixed in advance a rule for choosing one cell and a confirmatory test on fresh seeds, with a rewired control and a comparison with the published step period. No cell passes any of the four walking tests (rhythm: 0 cells), so the rule selects nothing, the stage stops, and its seven predictions fail. Adaptation does act: the tibia extensor pools fall from 21.3 Hz (weakest, fastest adaptation) to 0.55 Hz (strongest, slowest). But the flexor pools never exceed 0.22 Hz in any cell, so there is nothing for the extensors to alternate with. An exploratory diagnostic (not registered; exploration seeds) shows why. With every inhibitory connection removed the flexor pools fire at 151.341 Hz, so excitation does reach them. Of the neurons the walking command activates, each flexor motor neuron receives 274.73 inhibitory against 130.78 excitatory synapses, while each extensor motor neuron receives 754.17 excitatory against 698.17 inhibitory. GABA alone (182.68 per flexor neuron) outweighs that excitation. Whether the flexors stay net-inhibited if glutamate excites depends on the model's rule that every glutamate synapse inhibits (92.05 per flexor neuron). Under the model's signs, the walking command's population holds the flexors down and drives the extensors. A rhythm then needs something that periodically releases the flexors, which uniform adaptation does not provide. Candidates for a registered next step are the glutamate sign rule (glutamate excites through some receptors in the fly nerve cord), sensory feedback from the leg (stage 3), or a different intrinsic property (post-inhibitory rebound).

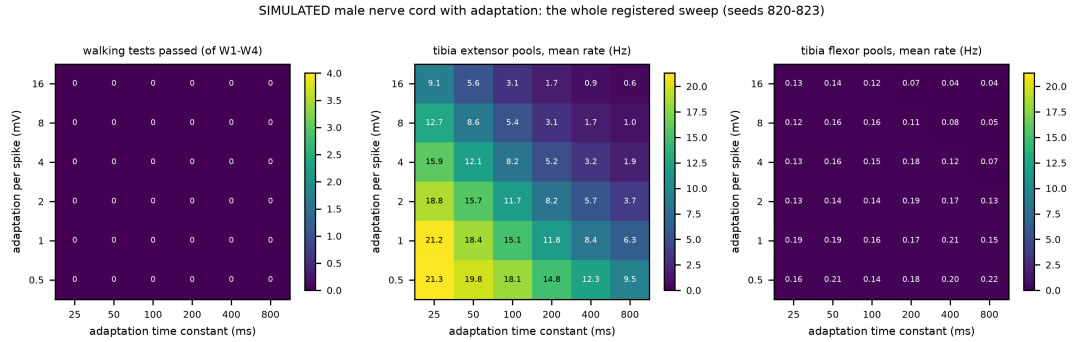


Figure 6. *The whole registered SIMULATED adaptation sweep: no cell passes a walking test; adaptation lowers the tibia extensor pools smoothly, while the flexor pools stay near zero everywhere.*

Stage 2d: the glutamate sign rule is not what keeps the cord from walking (1 of 7 at the selected cell). The model, like the published one, treats every glutamate synapse as inhibitory, and the stage-2c diagnostic showed that the flexors' net inhibition depends on that rule. No settled rule says where fly glutamate excites, so stage 2d registered two bounds. Glutamate onto motor neurons became excitatory in one; glutamate everywhere in the other. Each ran without adaptation and over stage 2c's grid (37 cells per rule), with the same selection rule and confirmation. Onto motor neurons only, the flexors stay weak (1.76 Hz without adaptation) and no cell passes any walking test. Everywhere, the flexors switch on (40.4 Hz), but the cord becomes far more active (7,118 neurons at 1 Hz or more, against 2,136 with glutamate onto motor neurons only). A rhythm appears in 5 cells, as slow waves of the whole population, but no cell alternates antagonists or legs (0 cells). The rule selected the one cell that passed the tripod test on exploration seeds; on fresh seeds that pass did not replicate. The left front flexor and extensor fire in phase there (distance 0.49 from the alternation target), the right front flexor stays silent, the rhythm's period is 317.5 ms, slower than a walking fly's, and only silence without drive passes. **Three registered attempts now agree.** The male nerve cord's wiring, run as leaky integrate-and-fire neurons under the walking command, does not produce alternation: not as published (2a), not with adaptation (2c), not with either glutamate rule (2d). This is a result for the paper. A connectome alone, in this model class, does not contain a working walking rhythm generator. The roadmap's own answer is the next stage, sensory feedback from the legs (stage 3), which the fly also needs to walk.

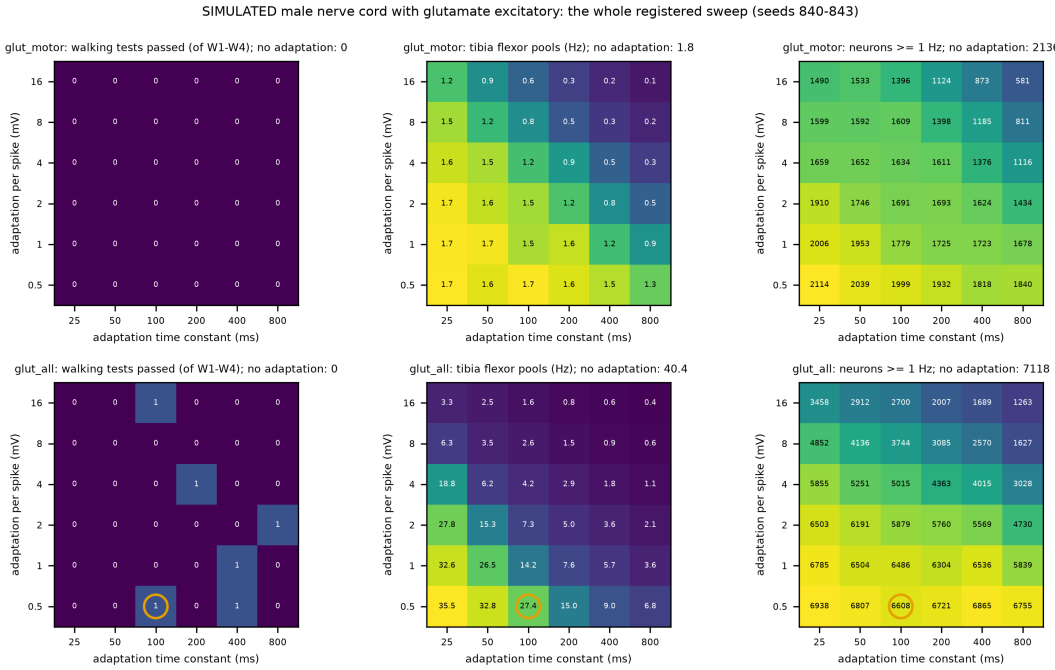


Figure 7. *The whole registered SIMULATED stage-2d sweep, both glutamate rules: walking tests passed, tibia flexor rate and the number of active neurons in every cell. Glutamate onto motor neurons leaves the flexors weak; glutamate everywhere turns them on with a large rise in activity, and no cell alternates.*

Stage 2e: the Pugliese et al. firing-rate model (2 of 8, one of them vacuous), and a flaw in our rhythm measure. Neither of the walking flies circulating online comes from a connectome. Eon Systems drives a NeuroMechFly walking controller trained by imitation, with a few descending neurons as handles. flybody (Google DeepMind with Janelia) walks with a trained policy. The one study that finds a walking rhythm in nerve-cord wiring is Pugliese et al. (2025). They use a firing-rate model with saturating, heterogeneous neurons. Steady drive to DNg100 produces leg rhythms through a three-neuron circuit (IN17A001, INXXX466, IN16B036). Stage 2e reimplemented that model from the paper and its configuration, ran it with their parameter distributions on the male nerve cord (32 draws), and registered eight predictions before any run. Silence without drive passes. Under DNg100 drive almost nothing happens: only 11 neurons become active, and the

rhythm predictions fail. The silenced-circuit prediction passes, but vacuously: there was no rhythm to remove. The cause is a stated deviation. The male CNS release has no neuron volumes, so size, which scales each neuron's threshold, was taken as total synapse count. That makes DNg100 one of the largest neurons in the cord (17.8 times the median, in the top 1 %, which starts at 11.86). Under the registered input it reaches only 6.54 Hz, against 49.32 Hz for a grooming neuron. So this run does not test Pugliese et al.'s result. Grooming and DNb08 drive, which reach their neurons, recruit about 7,055 neurons each. DNb08 scores a median rhythmicity of 0.7367, with a "period" of 50.0 ms in every draw, the shortest lag the measure looks at. The figure shows why. The left front flexor fires one burst and falls silent, and the extensor rises to a plateau; there is no rhythm. **The rhythmicity measure registered since stage 1d is flawed.** It takes the largest autocorrelation within lags of 50 to 500 ms without requiring a peak, so any smooth, slowly varying signal scores high at the 50 ms edge. Registered outcomes stay as registered, but three readings are now suspect:

- stage 1d's D2 (forward walking "swings" the leg; scored on the smooth claw distance);
- stage 2d's five "rhythmic" cells;
- every grooming comparison value.

The audit (registered before any corrected value was computed on stage data). The corrected measure counts a rhythm only at a genuine autocorrelation peak: a local maximum at least 0.1 above the trough it rises from. Its period is that peak's lag, and phases use that period. It was calibrated on synthetic signals only and passes its registered validation: smooth signals without a rhythm score 0, and a sine keeps its value. Stages 1d, 2a, 2c, 2d and 2e were re-simulated with their own code and seeds, and every trial was measured both ways. The registered measure reproduces every recorded outcome. Under the corrected measure:

- one registered outcome changes: stage 1d's D2 now fails (above);
- in stage 2d the cells with a rhythm drop from 5 to 1, and the selection rule would pick a different cell (glutamate everywhere, 16 mV, 100 ms); it is not confirmed here, since that would need fresh seeds;
- in stage 2e, DNb08's rhythmicity falls from 0.7367 to 0.0, and the share of rhythmic leg motor neurons from 0.699 to 0.01.

Stages 2a and 2c are unchanged. Registered outcomes stay recorded as they fell; this paragraph and the figure carry both. From now on, every new stage registers the corrected measure. Next, a registered stage-2e follow-up in which each command neuron gets enough drive to fire, so that Pugliese et al.'s result is actually tested.

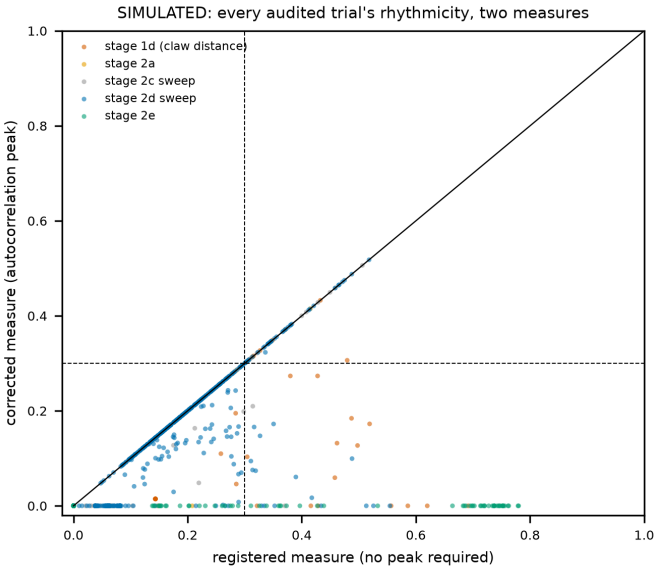


Figure 8. *SIMULATED: every audited trial's rhythmicity under the registered measure (no peak required) against the corrected one (an autocorrelation peak). Genuine rhythms lie on the diagonal; smooth signals the old measure scored as rhythmic fall to zero.*

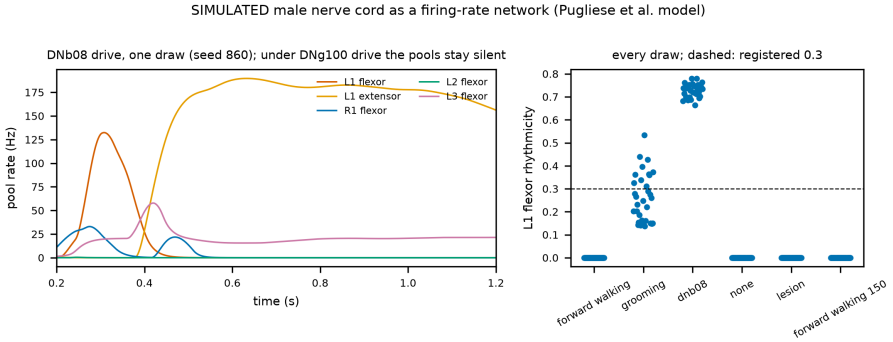


Figure 9. *SIMULATED firing-rate model of the male nerve cord. Left: DNb08 drive, one draw: one flexor burst, then a tonic extensor, scored as "rhythmic" by the flawed measure. Right: rhythmicity per draw and condition; DNg100 drive leaves the pools silent.*

Stage 2f: every command neuron driven alike; the walking command is switched off by the activity it starts (precondition failed in both size rules). Stage 2f gave each driven neuron the input that alone brings it to 50 Hz, computed from its own drawn parameters. It ran two size rules (synapse count, as in stage 2e, and none), used the audit's corrected rhythm measure, and added a precondition that DNg100 actually fires near its target. The precondition fails in both rules: DNg100's median rate is 0.18 Hz with synapse-count sizes and 0.0 Hz without size scaling, so every other prediction is uninterpretable. An exploratory trace of one registered draw (not scored) shows why. DNg100 starts at 42.67 Hz and recruits 443 neurons within 50 ms. The cord then tips into a self-sustained state, and by 0.25 s the inhibitory input onto DNg100 is -4452.5 against 2165.1 excitatory, which silences it. Thereafter 5,247 neurons stay active with the command off. The comparison with Pugliese et al. explains the difference. Their network is not the whole cord but the front leg's motor neurons, the premotor neurons onto them and the descending neurons onto those (4,604 neurons). Their input is halved whenever more than 500 neurons are recruited; such runs are treated as infeasible. Their positive result is a rhythm within the front leg, and they report no consistent left-right phase and no tripod. Our whole-cord runs sit far outside that regime, and our coordination tests ask for more than their model produces. The faithful test is their own construction on the male CNS: the front-leg network, the 5-to-500 recruitment rule, and their within-leg predictions. It is registered next as stage 2g.

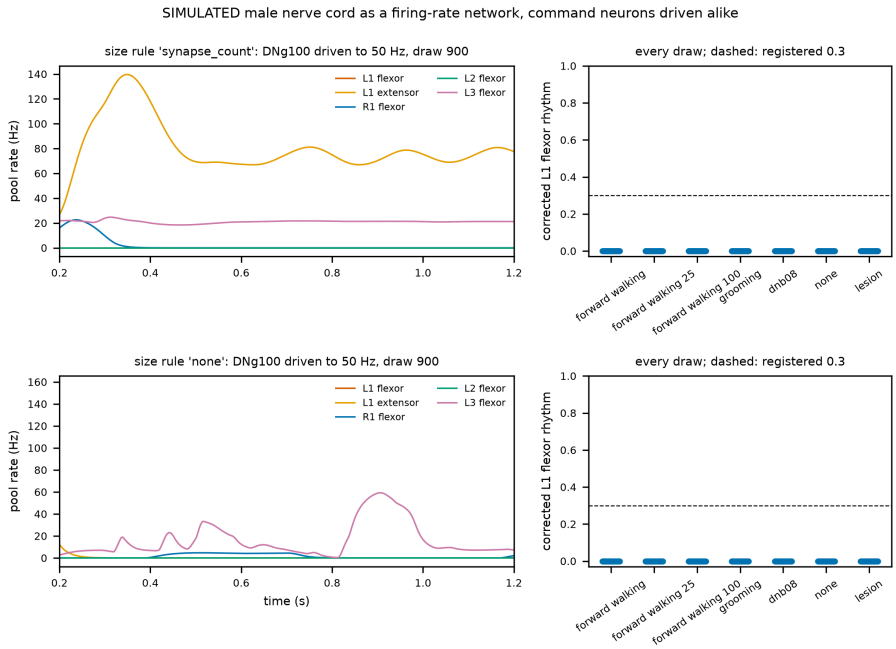


Figure 10. *SIMULATED firing-rate network of the male nerve cord with each command neuron driven to 50 Hz, both size rules: no corrected rhythm in any condition; the cord's own activity silences the walking command.*

Stage 2g: Pugliese et al.'s front-leg network on the male CNS (precondition failed narrowly; suggestive, not yet a test). Stage 2g rebuilt their network from the male CNS: the left front leg's motor neurons, the premotor neurons onto them and the descending neurons onto those (2,133 neurons). It used their recruitment rule: start at 250, halve the input above 500 recruited neurons, double it below 5, for up to 10 runs. Rhythm was scored with the corrected measure. The registered precondition, a working input in at least 16 of 32 draws, fails narrowly in both size rules: 12 draws with synapse-count sizes, 15 without size scaling. So every prediction is uninterpretable as registered. The rule fails because this network is close to all-or-none. In 20 draws (synapse-count sizes) and 11 (none), halving and doubling flip between almost nothing and about 1,044 recruited neurons, never landing between 5 and 500. What the feasible draws show is exploratory, not a test. A genuine, regular oscillation appears in the few motor neurons that are active (median 1.0 and 2.0, as their own "2-5 motor neurons"). Its corrected score is 0.8145 and 0.8747, with periods of 80.5 and 154.0 ms, within a walking fly's range. The rates are low (a few Hz). Without size scaling, silencing the three-neuron circuit removes the rhythm (median 0.0); with synapse-count sizes it does not (0.8145), so there the rhythm runs through another path. The tibia flexor and extensor pools are silent in these draws. The next registered step keeps their 5-500 criterion but searches the input finely (bisection between the last input that recruited too few and the first that recruited too many), on fresh draws. Then their within-leg result can be tested on every draw that has a working input at all.

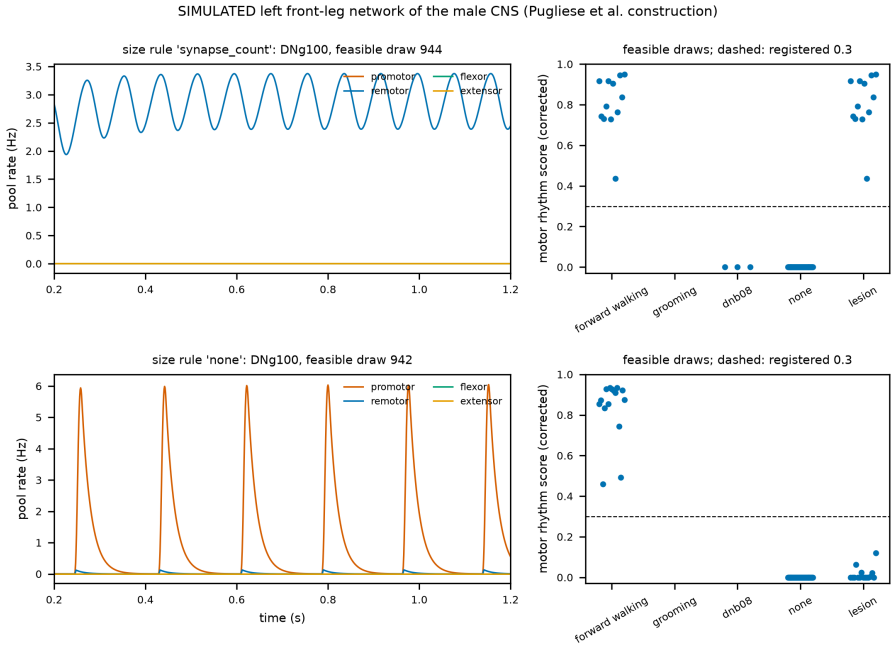


Figure 11. *SIMULATED left front-leg network of the male CNS (Pugliese et al.'s construction). Left: one feasible draw per size rule under DNg100: a regular low-rate oscillation in one or two motor-neuron pools. Right: corrected motor rhythm score over feasible draws, per condition.*

Stage 2h: with a fine input search, the walking command drives a front-leg rhythm. Without size scaling, that rhythm needs Pugliese et al.'s circuit; within-leg alternation still fails. Stage 2h kept stage 2g's network, criterion and predictions on fresh draws and changed only the input search. Once a draw had an input recruiting too few neurons and one recruiting too many, the next input was their geometric mean. A working input now exists in 32 of 32 draws with synapse-count sizes and in 18 without size scaling. In the draws without one, the bracket closes to a ratio of 1.0000001652591795: the network jumps from almost nothing to about 1,040 recruited neurons, a true discontinuity. Registered outcomes:

- the walking command drives a leg rhythm in both size rules (0.6808 and 0.8886), at a fly-like period (94.5 and 196.5 ms);

- silencing the three-neuron circuit removes it without size scaling (0.0) but not with synapse-count sizes (0.6808);
- no reliable promotor-remotor offset and no tibia alternation in either rule: the remotor and tibia pools are rarely active together with the promotor;
- the comparison with grooming "passes" vacuously in both rules, because no grooming draw found a working input and the code scored the empty group as zero.

The one assumption that could make the lesion result wrong is that IN17A001 is the only neuron receiving DNg100's input, so silencing it cuts the command's path rather than the rhythm. An exploratory check (not registered) settles this for the no-size rule. With IN17A001 kept and only INXXX466 and IN16B036 silenced, the network becomes more active (474.5 recruited neurons against 27.0), yet the rhythm is gone (0.0483). The rhythm sits mainly in the coxa promotor motor neurons (33 rhythmic instances). With synapse-count sizes no silencing variant changes anything: that rhythm runs through another path. **What this adds up to:** in an independent reimplementation on the male CNS, Pugliese et al.'s central claim holds without size scaling. A walking command drives a front-leg motor rhythm at a fly-like period, and the rhythm needs their inhibitory and excitatory interneurons, not just the relay. It depends on the size rule, and it does not give the antagonist alternation a walking leg needs. That is what our stage-2 question asked for and what Pugliese et al. also stop short of.

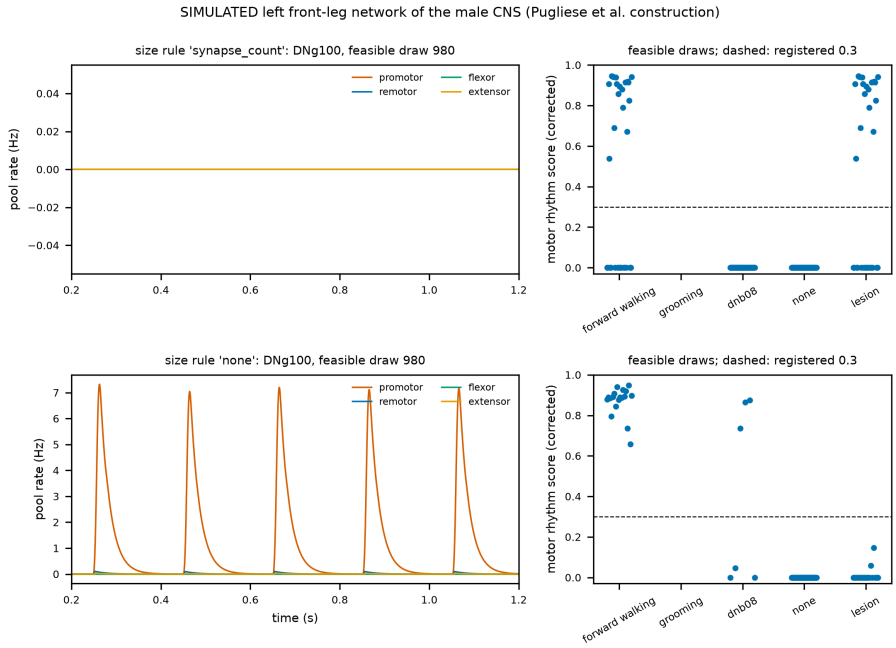


Figure 12. *SIMULATED left front-leg network of the male CNS with the fine input search. Left: one draw per size rule under DNg100 (no size scaling: regular promotor bursts about every 200 ms). Right: corrected motor rhythm score over feasible draws; without size scaling the lesion removes the rhythm.*

Stage 2i: with Pugliese et al.'s own size rule, neuron volume, the front-leg rhythm needs their circuit and promotor and remotor keep a fixed offset (6 of 7). Stages 2e-2h used stand-ins for neuron size because the male CNS flat files have no volumes. The project's neuPrint neuron table (same release and licence) does have them. It was fetched whole and checked against the bucket's CRC32C, and stage 2h was rerun unchanged with the published rule (size = volume over the network's median) on fresh draws. Every network neuron has a volume. Volume agrees only loosely with the synapse-count stand-in (Spearman 0.5738), and DNg100 is 11.1758 times the median. The walking command drives a regular front-leg rhythm (median score over the 32 feasible draws 0.9446, period 96.0 ms). Silencing the three-neuron circuit abolishes it (0.0115), which synapse-count sizes did not reproduce (stage 2h: 0.6808 with the circuit silenced). For the first time the promotor and remotor pools keep a reliable phase relation, defined in 27 of 32 draws with reliability 0.843. **The offset is not alternation, and that is the limit of the result.** Remotor peaks shortly before promotor in each cycle, and the two antagonists are never in antiphase. The tibia flexor pool stays silent, so flexor-extensor alternation fails again (V3). Reported without prediction: silencing the relay neuron IN17A001 alone leaves much of the rhythm (0.6575), and silencing INXXX466 and IN16B036 removes it (0.0263), as stage 2h found without size scaling. **What this changes:** with the published size rule, Pugliese et al.'s central front-leg claim reproduces on a second connectome, including the circuit's necessity. The synapse-count stand-in was the reason stage 2h's necessity test failed. Within-leg antagonist alternation is still not there.

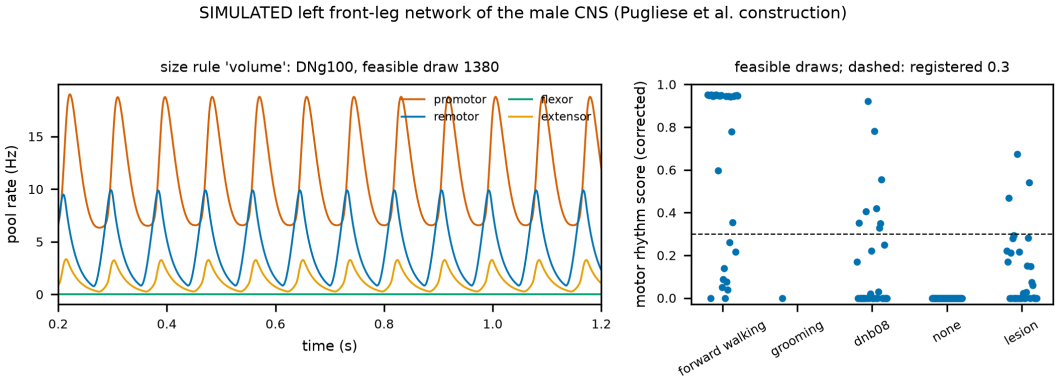


Figure 13. *SIMULATED left front-leg network of the male CNS with neuron volume as size. Left: one draw under DNg100: promotor, remotor and tibia extensor pools rise and fall together each cycle (remotor slightly ahead); the tibia flexor stays silent. Right: corrected motor rhythm score per feasible draw and condition.*

Stage 2j: the rhythm and its need for the circuit come from the neurons' own volumes; the fixed offset does not replicate (2 of 4). The assumption most likely to make stage 2i wrong was that any spread of sizes would do, or that the result hung on normalizing by the network's median. Stage 2j reran it on fresh draws in three arms. With the volume rule the rhythm and the circuit's necessity replicate (rhythm 0.6992, circuit silenced 0.0318). The promotor-remotor offset does not reach the registered reliability (R = 0.6082, needed 0.7), so J0 fails. With the same sizes shuffled across neurons, a fresh permutation per draw, there is no rhythm at all (0.0021), and the input search is all-or-none again: only 16 of 32 draws find a working input. J1 asked whether the circuit's necessity is lost under shuffling, and is scored fail as registered: with no rhythm there, necessity cannot be assessed. J2 passes: the offset is lost too. Normalizing by the nerve cord's median instead of the network's leaves everything in place, and the offset is tighter (rhythm 0.9531, R 0.9858, circuit silenced 0.0; J3 passes). **What this does to stage 2i:** the walking rhythm and its dependence on Pugliese et al.'s circuit need the identified neurons' own volumes, not just their spread. The promotor-remotor offset is real in some draws but not reliable across them at the registered line, so it is not claimed.

Stage 3a: the first closed loop. Claw-neuron feedback shifts the motor pools toward resistance but barely moves a held leg (go/no-go failed; 3 of 5). The male nerve cord and the muscle-driven left front leg now run together, one millisecond at a time. The leg's femur-tibia angle drives the four left front-leg claw neurons, the femoral chordotonal organ's position sensors, identified by type in the male CNS. The two types are wired differently: SNpp51 excite the tibia flexors, SNpp50 the extensor. Which type senses flexion is unknown, so a resistance assignment, its opposite and a non-contingent sham bracket it. Under the antennal-grooming drive, whose open-loop reach holds the tibia near full extension, resistance feedback moves the tibia by only 0.0054 rad, against the registered 0.1. The go/no-go fails, so the contingency test is uninterpretable. The loop stays silent without drive, the opposite assignment does not resist, and the reach survives with the loop closed (claw near the head for 0.968 of the second). At the motor level the feedback does act in the resisting direction: tibia extensor rates fall from 43.25 to 31.75 Hz and flexor rates rise from 0.1 to 3.4 Hz. But the sham does almost the same (33.25 and 3.0 Hz). In a held extension the claw neurons fire at a nearly constant rate (90.5232 Hz), so contingent and non-contingent feedback are the same input. An exploratory check (not registered) rules out muscle saturation as the reason the leg does not move. At a full-activation rate of 80 Hz, resistance moves the tibia from 0.6546 to 0.6705 rad and the sham to 0.6711. **What it means:** a static hold cannot show a reflex. The classic test, as in insect physiology, perturbs the joint and asks whether the loop pushes back faster than the same input delivered blind. That is the next registered step (stage 3b): a brief imposed flexion of the tibia during the held reach, with recovery time as the readout.

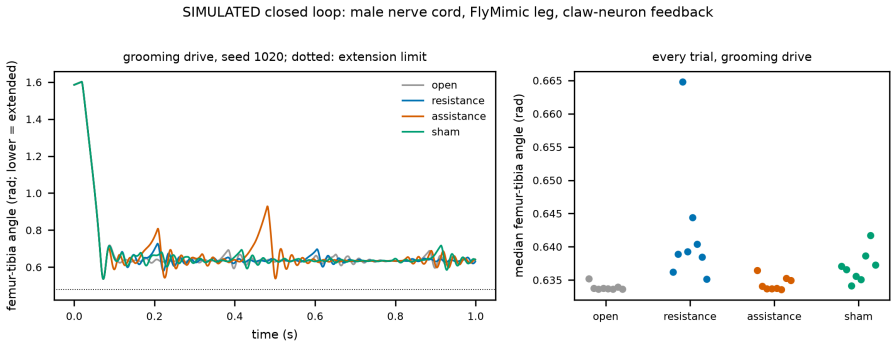


Figure 14. *SIMULATED closed loop (male nerve cord, FlyMimic leg, claw-neuron feedback), grooming drive. Left: femur-tibia angle over one trial per arm. Right: every trial's median angle; resistance and sham shift it by thousandths of a radian.*

Stage 3b: a push on the tibia. The claw-neuron loop does not return it faster; the leg's return is set by its muscles (2 of 5). During the held grooming reach a 20 ms torque flexed the tibia by a median 0.628 rad, so the precondition passes. The torque size came from a check on the leg alone. The sham kept the tonic claw feedback (the same seed's unperturbed resistance rates) and removed the part that follows the push. Resistance feedback did not speed the return. The leg came back 32.0 ms after the push, against 30.0 ms open loop, so the contingency test is uninterpretable. The opposite assignment was, if anything, slightly faster (27.5 ms), failing its prediction. The reach survives. In the traces the resistance and sham arms coincide: the part of the feedback that follows the push changes nothing measurable. The steady part acts instead. It lowers the extensor's rate (33.0 against 36.5 Hz) and so lets the push flex the leg further in some trials. **One data limit stands out.** The male CNS traces few proprioceptors for the front leg: 20 chordotonal neurons and 4 claw neurons on the left, against 89 and 22 for the left hind leg. A real femoral chordotonal organ has far more. A four-neuron pathway may simply be too thin to shape the movement. Stage 3's go/no-go has now failed twice. Feedback through this pathway changes the motor neurons but not how the front leg moves. Closing the loop where the sensors are well traced (the middle or hind legs) needs a muscle-driven body for those legs. FlyMimic models only the front leg, and flybody's legs are position-servoed.

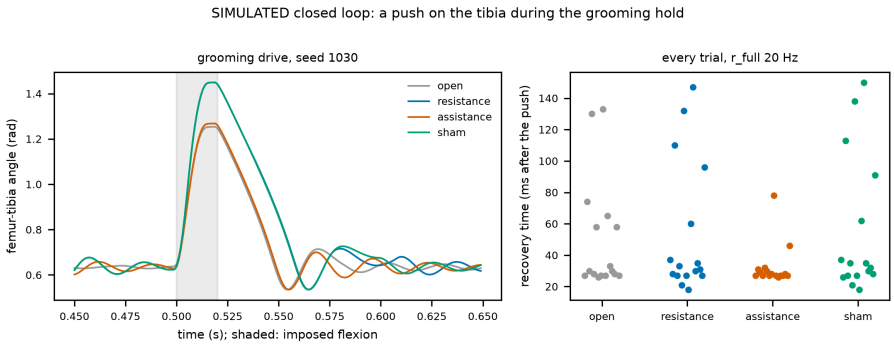


Figure 15. *SIMULATED push on the tibia during the grooming hold. Left: one seed per arm (the resistance trace lies under the sham). Right: every trial's recovery time; no arm returns faster than open loop.*

Stage 4a: drugs in the loop. With movement as the readout, the chip is still late (3 of 6). The registered chip-to-animal drug design was rerun with the validated movements as the behaviour: proboscis extension under sugar and under sugar with bitter, and the grooming reach through the brain-to-nerve-cord relay. Each transmitter-class knob acted on every network of a trial, and each family's movement onset was compared with the recorded chip and tract-electrode detections. Under vehicle both movements appear (proboscis 0.3044 mm, reach depth 0.9932 mm). Acetylcholine block stops both (-0.8944 and -0.8944). **The roadmap's question has a clear answer: the chip first flags every family at a higher dose than movement fails.** For example, GABA-A block impairs movement at 0.2 of the knob's range, and the chip first flags it at 0.6; GABA-A potentiation impairs movement at 0.5, the chip flags it at 1.0. Movement never fails earlier than the neural readouts of the parent study. It fails at the same dose in three families and later in two (excitability block 0.2 against 0.1): the body buffers small neural changes, but not enough to let the chip catch up. The output-tract electrode no longer leads: it flags in time in 4 families, the same as the planar array (4). The bitter gate is released by partial GABA-A block, as in the parent study. With bitter present the proboscis extends severalfold more than under vehicle (0.0671 mm; figure), but at full block every movement collapses. The relation is therefore not monotonic and its rank-correlation prediction fails (-0.0286). One scoring detail is reported, not hidden. At full GABA-A block the claw ends beyond its resting distance (depth -0.0146 mm), so that condition's disruption index averages only the two proboscis terms. It does not change any onset. **What this adds to the chip-to-animal result:** the chip's lateness, found at the neural and motor-neuron levels, holds for the movements themselves. A drug that the simulated chip first flags at a given dose has already changed how the simulated fly feeds and grooms at a lower one.

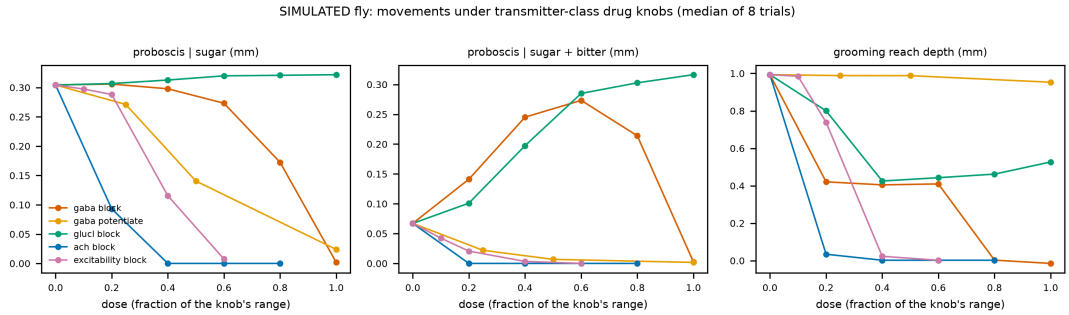


Figure 16. *SIMULATED fly movements under the five transmitter-class drug knobs (median of 8 trials per condition): proboscis extension under sugar and under sugar with bitter, and the depth of the grooming reach.*

Stage 4b: the headline on fresh seeds. The chip is still late in every family (3 of 4). Stage 4a rested on one set of 8 seeds and on the chip detections of the parent's first run, so it was rerun unchanged on 16 fresh seeds and compared with the chip detections of the parent's own fresh-seed replication. Each movement was also left out of the disruption index in turn. The movements appear under vehicle (proboscis 0.304 mm, reach depth 0.9893 mm). In all five families movement fails first and the chip flags later (R1 passes): GABA-A block at 0.2 against 0.6, GABA-A potentiation at 0.5 against 1.0. Every movement onset is within one dose step of stage 4a's (R3 passes). Only excitability block moved, from 0.2 to 0.4. The leave-one-out check (R2) fails in one family. Without the bitter-gated proboscis, GABA-A potentiation impairs movement only at 1.0, the dose at which the chip flags it, so the two tie. At the dose below, the proboscis under sugar extends about half as far as under vehicle, but the reach does not change, so the two-movement index stays just below the line (`outputs/fly/embodied_stage4b_SIMULATED.json`, medians). Two further readings are reported, not scored. Against the chip's first run rather than its replication, excitability block would tie: both at 0.4. Stage 4a's bitter-gate correlation comes out the same on the fresh seeds (-0.0286). As in stage 4a, at full GABA-A block the claw ends beyond rest (depth -0.0146 mm), and that index averages the two proboscis terms. **What this does to the headline:** the chip's lateness at the movement level replicates on new seeds and against the chip's own replication. In four of the five families it does not depend on any one movement. In GABA-A potentiation it rests on the bitter-gated proboscis, and in excitability block it is one dose step, so in two families the margin is a single step of the dose grid.

Stage 4c: inside the one-step margins (3 of 4). Stage 4b's two thin margins were rerun on a finer dose grid: GABA-A potentiation at four levels between its two highest doses, excitability block at four levels on each side of its movement onset. The movements used stage 4b's seeds and the chip both of the parent's seed sets. The coarse levels, rerun alongside, reproduce their recorded values to floating-point rounding (F0 passes). Excitability block resolves the way the headline says, against either chip run. Movement fails at 0.24, and the chip first flags at 0.36 (replication) or 0.4 (first run). In GABA-A potentiation the finer grid brings the chip forward, to 0.6, still after movement fails at 0.5. Without the bitter-gated proboscis the two stay tied, now at 0.6 on the fine grid (F2 fails). **What this does to the headline:** at the movement level, in this model, the chip is late in every family with the full index. In one family, GABA-A potentiation, that lateness depends on the proboscis movement that bitter gates; without it, chip and movement change together.

Stage 4d: walking as a drug readout. The chip is late for walking under GABA-A and GluCl block, early under acetylcholine block, and GABA-A potentiation does not touch walking (2 of 4). The drug knobs were applied to stage 2i's walking network (rate model, volume sizes) on fresh draws: the transmitter knobs scale weights by presynaptic class, as in the spiking model, and the excitability knob scales each neuron's activation threshold. Each draw's DNg100 input was found under vehicle and held fixed. Under vehicle the walking command drives the rhythm (median score 0.8222; W0 passes). GABA-A block and GluCl block break it at the lowest dose (0.2 and 0.2), long before the chip flags them. **Acetylcholine block is the first case anywhere in this study where the chip flags first**: the chip at 0.4, walking at 0.6 (W1 fails). The rhythm survives milder block intact and then collapses. GABA-A potentiation leaves walking as regular as under vehicle or more so (median score 0.947 at the top dose), so W2 fails: the lateness in that family comes from feeding alone. The excitability knob barely changes walking either. With walking added as a fourth movement, the chip is still late in all five families (W3 passes), because the other movements fail first where walking does not. One reading caution: within a condition the draws are bimodal, strongly rhythmic or not, and under vehicle about half are rhythmic, so the median sits between the modes and mild knobs that tip a few draws raise it (`outputs/fly/embodied_stage4d_SIMULATED.csv`). **What this changes:** the headline holds for the movement index. It does not hold for every readout, and which behaviour is read decides which drugs a chip catches in time.

Stage 4e: the acetylcholine exception does not survive a readout that respects the draws (2 of 4). Stage 4d's draws were bimodal, so the registered retest scored walking by the fraction of draws with a rhythm (score at least 0.3), on fresh draws. Under vehicle 0.5938 of draws are rhythmic (E0 passes). GABA-A and GluCl block break walking before the chip flags them (E2 passes). Acetylcholine block does not: at its mildest dose nearly every draw becomes rhythmic, and the two-sided index, used for every movement in this study, counts that rise as a change at 0.2, before the chip's 0.4 (E1 fails). GABA-A potentiation likewise raises the rhythm and counts as a change at 0.25 (E3 fails). Scored by stage 4d's median on the same fresh draws (reported, not tested), acetylcholine block is again chip-first (walking at 0.6). **What this changes:** stage 4d's exception depends on the readout, so it is withdrawn from the summary claims. The robust walking result is that GABA-A and GluCl block break the rhythm long before the chip flags them. The walking network's vehicle state sits near a transition, so mild knobs that strengthen it tip draws into rhythm. That makes a fictive rhythm read on its own a fragile drug readout in this model.

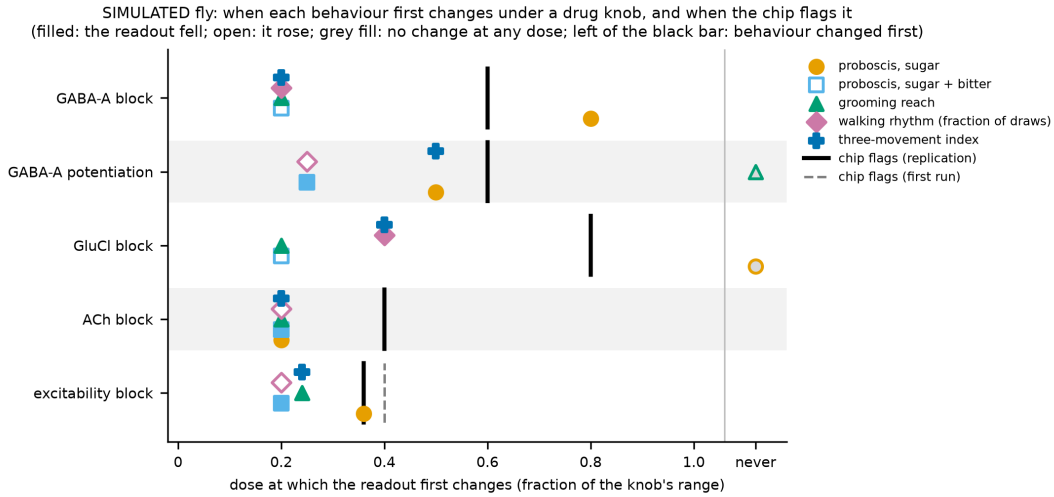


Figure 17. *SIMULATED drug onsets of stages 4b-4e in one view (`fly.onset_figure`, recorded outputs only): per drug family, the dose at which each movement, the walking rhythm (fraction of draws) and the three-movement index first change, filled where the readout fell and open where it rose, against the dose at which the chip first flags (black: replication; dashed: first run, hidden where the two agree).*

Stage 5: checking directions against real flies. The registered search found no fly pharmacology to compare with (both predictions uninterpretable). Before any search, the stage registered ten queries, inclusion rules, a mapping from published manipulations to the knobs and a rule for reading a simulated direction from stage 4a. This guards against choosing, after the fact, the papers that agree. The searches found no study that gives an adult fly a GABA-A, GluCl, nicotinic or sodium-channel drug, or manipulates those receptors genetically, and scores proboscis extension or grooming: 0 qualifying comparisons, so both registered predictions are uninterpretable. The closest receptor study (Rdl knockdown raises consumption; Cheung and Scott 2017) measured consumption, which the registration placed out of scope. Studies that silence or activate identified GABAergic neurons give 3 comparisons, reported beside as registered, and the simulation agrees with 1. Silencing the GABAergic 13A/13B leg neurons reduces anterior grooming (Syed, Ravbar and Simpson), as GABA block reduces the simulated reach. Activating them also reduces grooming, while GABA potentiation leaves the simulated reach unchanged. Silencing four GABAergic interneurons raises proboscis extension to sugar in fed flies (Pool et al. 2014), while GABA block lowers it in the simulation. Two differences explain the mismatches, and they are the model's to state. A transmitter-class knob changes every synapse of that class, not a few identified neurons. And the simulated fly has no satiety: its sugar response is already near full extension, while a fed fly's is low and has room to rise. The search also recorded 16 excluded candidates, 2 findings the model cannot be compared with (a GABA-B result and responses to bitter or water alone), and 4 pages that could not be read; all are listed in `src/neuromps/fly/real_fly_literature.json`. **What stage 5 shows:** the simulated drug

effects cannot yet be checked against fly pharmacology on these behaviours, because the experiments have not been published. That gap is itself a result: it marks the experiments that would test the model.

Stage 6: release. One command reproduces every fly number. `neuromps.fly.pipeline` reruns every fly study after the chip-to-animal study in dependency order, each as its own process, and `python run_all.py --full` runs it as its `embodied` stage. Run from scratch, the stage took 10884.5 s on the GB10 and reproduced every recorded value: the only differences were re-verification timestamps in the data manifest, the order of keys in two output files, and one field later code adds. Two overnight additions run in the same pipeline. First, the movement-level analogue of the precision table (exploratory): at the dose where movement first fails, the chip has moved by only 0.149 log2 under GABA-A block and at most 0.452 under GABA-A potentiation, which is within the noise of real chips (the noisier recorded floor is 1.752). Second, the cell the corrected rhythm measure selects in stage 2d (glutamate excitatory everywhere, 16 mV, 100 ms), confirmed on fresh seeds as the audit left open: its rhythm does not replicate (0.2035 under the walking command against 0.3 required), so the glutamate line stays closed. Every choice of path is recorded, with its alternatives, in `report/DECISION_LOG.md`.

Stage 5b: the identified GABAergic leg neurons, manipulated as the experiment did (2 of 5). Stage 5 could compare the published identified-neuron results only with class-wide knobs. Stage 5b makes the published manipulation instead, in the grooming relay of stage 1f: the left front leg's GABAergic 13A or 13B hemilineage neurons (71 and 73 neurons in the nerve-cord network) were silenced, their outputs removed, or activated with a 150 Hz drive. The published direction is less grooming under both (Syed, Ravbar and Simpson). Silencing 13A reproduces it: the reach falls from 0.9927 to 0.3806 mm. The other three do not. Silencing or activating 13B leaves the reach unchanged (0.9923 and 0.9927 mm). Activating 13A almost silences the front-leg motor pool (1.1691 Hz against 13.9265 intact) but shortens the reach only to 0.7309 mm, short of the registered line. Two readings of the mismatches stand out. The experiment manipulated sub-populations and read the joints of a grooming leg, while the model manipulates whole hemilineages and reads one reach. And 13B's outputs in this network appear not to reach the muscles that carry the reach. Both are listed as branches in `report/DECISION_LOG.md` (D17).

Stage 5c: the same manipulations read at the joint. None of the four published postures appears (0 of 4), and stage 5b's one agreement came by another route. The study stage 5b compares with read the joints of the front legs: silencing 13A or 13B locks them in flexion, and continuous activation over-extends them. Read at the femur-tibia joint on fresh seeds, the model does otherwise. Silencing 13A extends the tibia (-0.1488 rad), so the shorter reach of stage 5b did not come from a flexion lock. Silencing 13B flexes it by only 0.082 rad. Activating 13A lets the leg fall back toward its passive rest (+0.8325 rad) instead of over-extending it. In this model the inhibitory hemilineages act as broad brakes on the leg's motor neurons, not as the joint-specific controllers the experiment shows: a limit of a connectome-only, leaky integrate-and-fire nerve cord to state beside the stage-2 walking results.

Stage 3c: the perturbation test on a well-traced leg. The feedback acts, but it makes the hold unstable (2 of 4). Stages 3a and 3b used the front leg's four claw neurons. Stage 3c closed the loop through the right hind leg's 25, with flybody's servoed tibia: the walking command holds the tibia extended, and a 20 ms torque pushes it. The precondition passes (held at 1.0284 rad, pushed by 0.4952 rad), and the opposite assignment does not speed the return. Resistance feedback does not speed it either (23.0 ms against 19.0 ms open loop), so the contingency test is uninterpretable as registered. The reason differs from the front leg. Here the feedback is strong: it lowers the extensor pool from 34.0 to 18.0 Hz, the held angle falls to 0.7182 rad, and the traces show the leg swinging irregularly rather than holding. Against the sham, which receives the same tonic feedback, the resistance arm returns sooner (23.0 against 32.5 ms; exploratory, with recovery times spread widely in this unstable regime). So the feedback that follows the push might act on this leg, unlike the front leg. **Stage 3d tested that difference alone, paired by seed, on 32 fresh seeds, and it does not hold (0 of 2).** The resistance arm returns sooner than the sham in 14 of 32 seeds (22 needed), and the median paired difference is 0.0 ms. Closed through either leg, the claw neurons' feedback changes the motor neurons and, on the hind leg, the posture, but no part of it that follows a push could be shown to speed the leg's return. Stage 3's go/no-go stays failed; what would change it is a muscle-driven, well-traced leg, which no available body model provides.

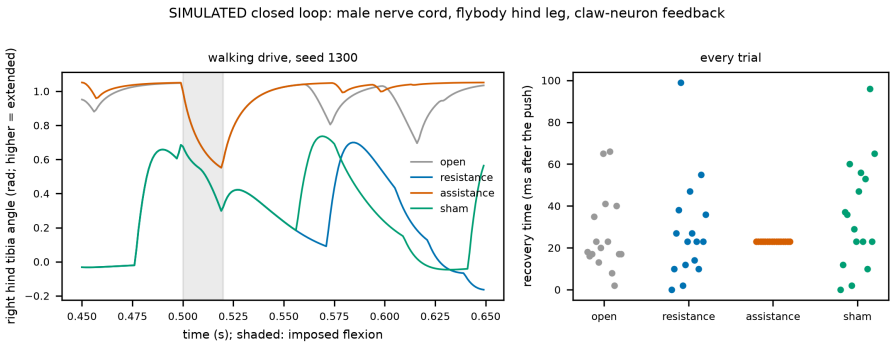


Figure 18. *SIMULATED push on the right hind tibia (flybody, servoed) with claw-neuron feedback. Left: one seed per arm (the resistance trace lies under the sham until about 0.55 s); with resistance feedback the leg swings irregularly instead of holding. Right: every trial's recovery time.*

8. What carries over unchanged

Pre-registration before every run; licensed data with checksums in the manifest; every reported number written by a script and checked against the documents that cite it; failed predictions reported beside the passed ones; and a demo anyone can open.